



MiniZinc Challenge 2025; 3 reps per (mode, instance), n = 70 of 100 shown (the remaining 30 had at least one rep that hit the 1200 s wall-clock timeout and were excluded so the parsed internal times aren't truncated). Times shown are solver-reported “internal” time — the last MiniZinc-emitted time field for standalone and the last % time elapsed: line for parasol — both measured from process start to the final event before shutdown. That excludes parasol’s post-event shutdown tail (median ≈ 50 ms in this sweep), so the delta isolates pure orchestration overhead rather than total wallclock. Each point is one instance: x = median standalone-internal time over 3 reps; y = median(parasol) — median(standalone). Error bars span the worst-case delta given the observed per-mode rep ranges (min-parasol — max-standalone, max-parasol — min-standalone), so their size reflects cp-sat’s own run-to-run variance with 8-core parallel search and stochastic restarts. Median overhead across these 70 cleanly-finished pairs is 0.171 s (mean 6.613 s, stdev 56.526 s). The within-mode noise floor — median range across the 3 reps — is 0.472 s for standalone and 0.558 s for parasol, i.e. larger than the median overhead, so for any single instance the delta you’d measure from one rep is dominated by solver variance, not by parasol’s per-invocation cost.